

Chapter -1

Global Steel Scenario & Indian Steel Industry

1.1 Introduction

Though Iron and steel have been used by men for almost 6000 years, yet the modern form of iron and steel industry came into being only during the 19th century. The growth and development of iron and steel industry in the world until the Second World War was comparatively slower. But the industry has grown very rapidly after the Second World War. By 1970, world steel production had reached 595 MT from 189 MT in 1950. However during 1970-2000, rate of growth of steel production somewhat slowed down due to static steel demand in the developed economies of USA, Europe, Japan and USSR and various geo-political crises such as the oil crisis of 1970s, Afghanistan war of 1980s and Asian currency crisis of 1997.

World Steel production gained further momentum due to rise in production and demand in Asia particularly in China. In 1993, China overtook US and in 1996, it overtook Japan, as the top producer of steel. Since 2000, world steel production has grown rapidly on account of explosive growth of steel production in China. From a production figure of just 131 MT in 2000, China's crude steel production rose to 1,013 MT in 2022 which was 54% of total world steel production. Steel production temporarily dipped in 2008 due to global financial crisis and in 2016 due to production clampdown by China. But otherwise Chinese steel production since 2000s have seen a steady growth rate.

Global steelmaking capacity reached approximately 2.55 billion metric tons by the end of 2025. The Organization for Economic Co-operation and Development (OECD) has warned that global steel overcapacity is set to exceed 680 million mt by the end of 2025, marking the fastest expansion since the 2009 financial crisis. After three years of contraction (2022-2024), global steel demand is stabilizing in 2025 and may recover by one percent in 2026, returning to 2022 levels. While demand in China continues to decline sharply, India, ASEAN countries and other developing economies are expected to sustain growth, partially offsetting the downturn in developed markets.

Historical Background

There are evidences that man knew the use of iron since the ancient civilization of Babylon, Mexico, Egypt, China, India, Greece and Rome. Archeological findings in Mesopotamia and Egypt have proved that iron or steel has been in the service of mankind for nearly 6000 years. The origin of the methods used by early man for extracting iron from its ores is unknown.

In early days the iron produced probably was so relatively soft and unpredictable, that bronze continued to be preferred for many tools and weapons. Eventually iron replaced the non-ferrous metal for these purposes when man learned how to master the difficult arts of smelting, forging, hardening and tempering iron.

In the beginning, iron was smelted by charcoal made from wood. Later coal was discovered as a great source of heat. Subsequently, it was converted into coke, which was found to be ideal for smelting of iron. Iron kept its dominant place for 200 or more years after the Saugas works the first successful Iron Works in America, was founded in 1646. About the mid-19th century the new age of steel began with the invention of Bessemer process (1856) making steel available in large quantities at reasonable cost.

Indian History

Indian history is also replete with references to the usage of iron and steel. Some of the ancient monuments like the famous iron pillar near New Delhi or the massive beams used in the Sun Temple at Konark bear ample testimony to the technological excellence of the Indian metallurgists.

The history of iron in India goes back to the ancient era. Our ancient literary sources like Rig Veda, the Atharva Veda, the Puranas and other Epics are full of references to iron and to its uses in peace and war. According to one of the studies, iron has been produced in India for over 3000 years, in primitive, small scale facilities.

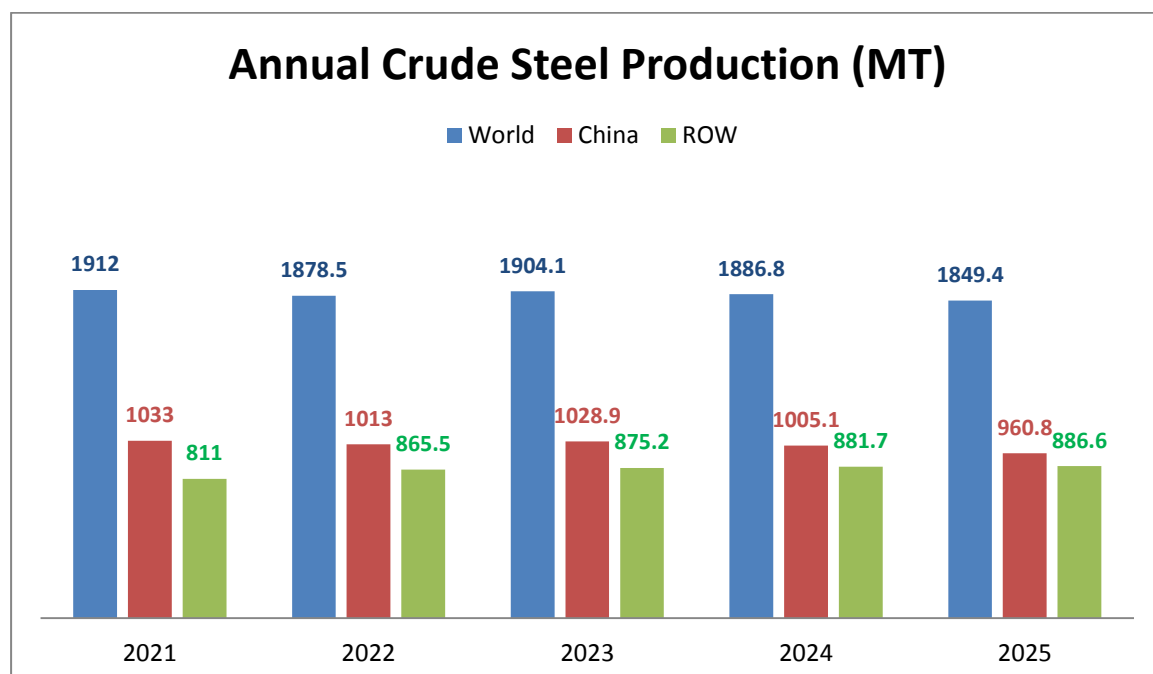
1.2 Global Scenario

Crude steel production 2025 (WSA)

Total world crude steel production was 1849.4 MT in 2025, down by 2 % over 2024. India was the second largest producer of steel in the world in 2025. List of top ten producers of steel in the year 2022 is given below:

Rank	Country	CY 2025	CY 2024	% change 2025/2024
1	China	960.8	1 005.1	-4.4
2	India	164.9	149.4	10.4
3	United States	82.0	79.5	3.1
4	Japan	80.7	84.0	-4.0
5	Russia (e)	67.8	71.0	-4.5
6	South Korea	61.9	63.6	-2.8
7	Türkiye	38.1	36.9	3.3
8	Germany	34.1	37.3	-8.6
9	Brazil	33.3	33.9	-1.6
10	Iran (e)	31.8	31.4	1.4

Total Production of Crude steel; from 2021 to 2025-World, China, Rest of World (RoW)



Source: WSA

World Steel Demand (WSA)

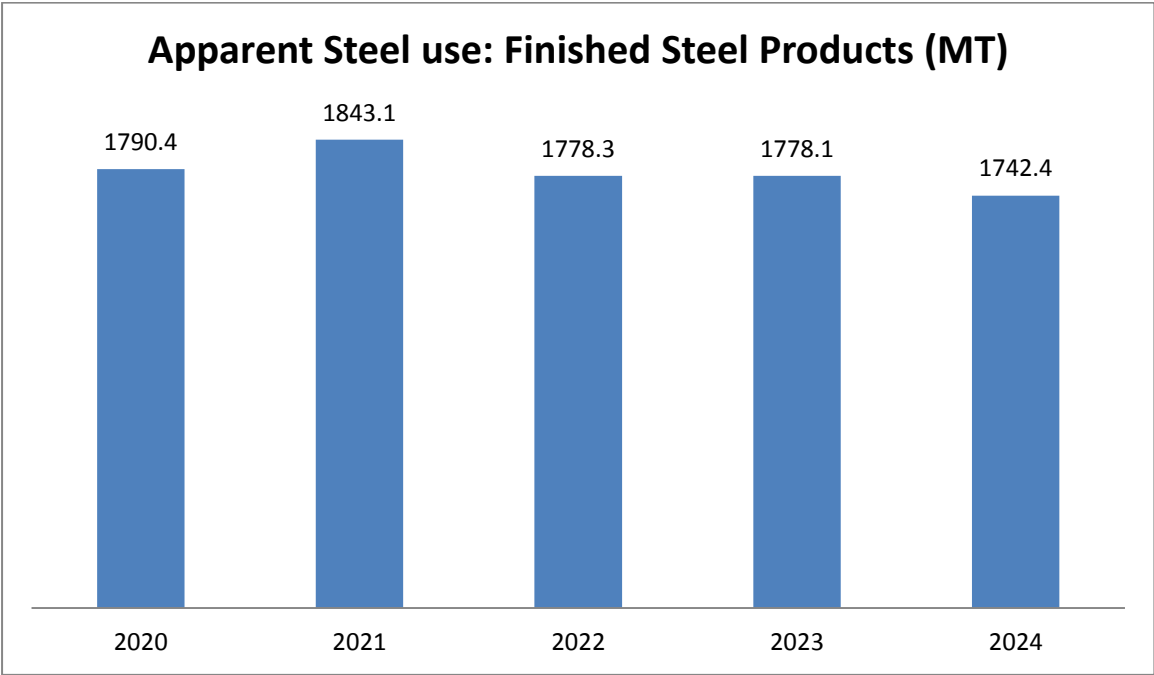
Regions	million tonnes			y-o-y growth rates, %		
	2024	2025 (f)	2026 (f)	2024	2025 (f)	2026 (f)
European Union (27) & United Kingdom	140.3	142.0	146.6	2.0	1.3	3.2
Other Europe	45.3	46.9	47.7	1.5	3.4	1.8
Russia & other CIS + Ukraine	59.2	56.1	55.2	-1.9	-5.2	-1.7
USMCA	129.7	128.1	130.9	-2.1	-1.2	2.2
Central and South America	46.8	49.4	50.4	2.2	5.5	2.2
Africa	40.1	41.1	43.0	5.8	2.4	4.7
Middle East	57.9	60.1	62.5	5.5	3.7	4.0
Asia and Oceania	1 230.0	1 225.5	1 236.1	-2.7	-0.4	0.9
World	1 749.4	1 749.2	1 772.5	-1.6	0.0	1.3
World excl. China	892.8	909.7	941.4	2.3	1.9	3.5
Developed Economies	348.9	347.2	352.5	-2.1	-0.5	1.5
China	856.6	839.5	831.1	-5.4	-2.0	-1.0
India	147.9	161.1	175.7	11.4	8.9	9.1
Em. and Dev. Economies excl. China & India	395.9	401.4	413.3	3.4	1.4	3.0
ASEAN (5)	78.5	80.8	84.1	8.4	3.0	4.0
MENA	74.5	76.9	80.2	8.1	3.2	4.4

Fig:- World Steel Demand

Source: Worldsteel Short range outlook Oct'25

Global steel demand in 2025 is projected to be flat compared to 2024, reaching about 1,749 million tonnes (Mt). A modest rebound of 1.3% is forecast for 2026, pushing global demand to 1,773 Mt. It is expected to see a slowdown in the decline of steel demand from China, coupled with strong growth in developing economies like India, Vietnam, Egypt, and Saudi Arabia. Long-awaited return of steel demand growth in Europe is also anticipated. Steel demand in the developing world

excluding China is forecast for robust growth, with a 3.4% increase in 2025 and a 4.7% increase in 2026. This expansion is primarily driven by strong performance in India, and some ASEAN and MENA countries. For nearly a decade, starting in the mid-2010s, steel demand in Africa remained largely flat, hovering around the 35–40 Mt mark. However, a significant shift has been underway since 2023, with clear indicators of a strong resurgence in construction and steel consumption across the continent. WSA analysis estimates that over the past three years, Africa’s steel demand has grown by an average of 5.5% per annum, fueled particularly by robust activity in the Northern and Eastern regions. This renewed momentum, which brought Africa’s steel demand to about 41 Mt in 2025, is underpinned by improving macroeconomic fundamentals and governance.



Source: WSA (World Steel in Figures 2025)

1.3 Growth of Indian Steel Sector

Steelmaking capacity in India has been expanding rapidly in recent years, and the country is now ranked second after China in terms of crude steel production. Further growth is expected in the medium to long term. The growth in the Indian steel sector has been driven by rising domestic demand for steel, low per capita steel consumption, domestic availability of raw materials such as iron ore and cost-effective labour. Consequently, the steel sector has been a major contributor to India’s manufacturing output.

India was the world’s second largest steel producer in CY’2025 with a production figure of 164.9 MT, up by 10.4 % y-on-y. In 2025, India’s crude steel production capacity reached 200 MT & India has planned to reach 300 million tonnes per annum (mtpa) of crude steel capacity by 2030.

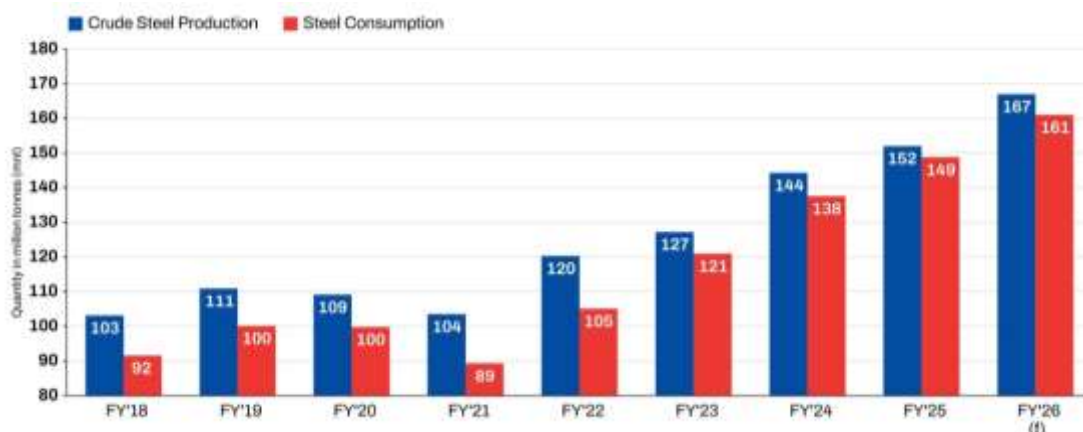


Fig: India Crude Steel Production & Consumption trend

Source: Bigmint

India's crude steel production and consumption show a clear structural uptrend despite the FY21 pandemic dip. Production rose from 103 mt in FY18 to a projected 167 mt in FY26, while consumption increased from 92 mt to 161 mt in the same period. The sharp rebound post-FY21 highlights strong domestic demand recovery, driven by infrastructure, construction, and manufacturing growth. Notably, the gap between production and consumption has narrowed in recent years, indicating robust internal absorption and reduced reliance on exports for balance. Overall, the trend reflects India's steady positioning as a high-growth steel market with strong demand fundamentals.

Item	Performance of Indian steel industry		
	April-March 2024-25*(MnT)	April-March 2023-24(MnT)	% change*
Crude Steel Production	151.967	144.299	5.3
Hot Metal Production	91.339	87.045	4.9
Pig Iron Production	8.334	7.364	13.2
Sponge Iron Production	55.654	51.560	7.9
Finished Steel (alloy/stainless + non-alloy)			
Production	146.560	139.151	5.3
Import	9.551	8.320	14.8
Export	4.858	7.487	-35.1
Consumption	152.001	136.290	11.5
Source: JPC; *provisional; MnT=million tonnes			

India's steel industry delivered broad-based growth in FY25, with crude steel production up 5.3% and finished steel production rising similarly. The standout trend is demand strength; finished steel consumption surged 11.5%, significantly outpacing production growth. Imports jumped 14.8% while exports fell sharply by 35.1%, indicating stronger domestic absorption. Overall, the data reflects a demand-driven expansion phase with increasing focus on the domestic market.

Period	Finished Steel Production (MT)	Import (MT)	Export (MT)	Consumption (MT)
2019-20	102.6	6.8	8.4	100.2
2020-21	96.2	4.8	10.8	94.9
2021-22	113.6	4.7	13.5	105.8
2022-23	123.2	6	6.7	119.9
2023-24	139.2	8.3	7.5	136.3
2024-25	146.6	9.6	4.9	152

Source: JPC

Market Size

India's crude steel production capacity reached 200 MT in 2025

Producer wise overall Market Share						
Producer	Apr-March, 2024-25		23-24	22-23	21-22	20-21
	Quantity ('000 T)	% Share	% Share	% Share	% Share	% Share
SAIL	16121	10.6	11.1	12.3	12.4	12.3
Tata	20624	13.6	14.2	14.6	15.6	11.3
JSW Steel	21064	13.9	14.3	14.8	11.3	11.5
RINL	3215	2.1	2.7	2.9	3.5	3.0
AMNS	6767	4.5	4.8	4.9	5.4	5.9
JSPL	7040	4.6	4.0	4.4	4.4	4.8
Other Producers	67618	44.5	42.9	41.1	43.1	47.6
Imports	9550	6.3	6.0	5.0	4.3	3.7
Total	152000	100.0	100.0	100.0	100.0	100.0

Source: MAG

The market share data for FY25 indicates increasing fragmentation alongside rising import penetration. JSW Steel (13.9%) and Tata Steel (13.6%) remain market leaders, though both have seen gradual moderation in share compared to earlier years. SAIL's share has also declined to 10.6%, reflecting intensifying competition. Notably, "Other Producers" command a significant 44.5%, highlighting a highly diversified market structure. Imports have steadily risen to 6.3% share, underscoring growing external competition amid strong domestic demand.

Unlike other large steel producers, the Indian steel industry is also characterized by the presence of a large number of small steel producers who utilize sponge iron, melting scrap and non-coking coal (EAF/IF route) for steelmaking.

Particulars (India)	FY 2025	FY 2024
% Share of oxygen Route production	41	43
% Share of EAF Route production	21	22
% Share of IF Route Production	38	35

Source: JPC

India's steel production mix in FY25 shows a gradual shift toward the induction furnace (IF) route, whose share increased to 38% from 35% in FY24. Meanwhile, the oxygen (BF-BOF) route declined to 41% from 43%, and the EAF route saw a marginal drop to 21% from 22%. This indicates a slight tilt toward secondary steelmaking, likely driven by cost dynamics, scrap availability, and strong long-product demand, while the primary route still retains the largest share in overall production. However, the large players of Indian Steel Sector (SAIL, Tata, JSW, AMNS, JSPL) have chosen BF-BOF route for their expansion plans.

Outlook for Indian Steel industry

For steelmakers, India is emerging as a key market with the potential to lead the way in future growth. Backed by resilient domestic demand, large-scale infrastructure investments, and favorable policy tailwinds, India's trajectory could reshape the steel narrative.

Structural drivers point to an improved demand outlook. India's steel consumption is expected to grow at a 5 to 6 percent CAGR, rising to reach between 240 million and 260 million metric tons by 2035. Growth in the sector is expected to be driven by GDP expansion, a \$1.4 trillion National Infrastructure Pipeline, and rising consumption from the renewable energy and defense sectors.

However, domestic steel prices started rising from December onwards supported by rising international steel prices and better performance by domestic end-user industries. Major steel players have announced capacity additions of approximately 70 million metric tons per annum over the next five to ten years, driving much of this growth. Most of these additions are expected to support blast furnace–basic oxygen furnace (BF-BOF) processes, which could increase lock-in risks for carbon-intensive assets.

India's demand for steel is expected to be driven by the demand for high-grade steel from sectors such as automotive, renewables, and defense, which is accelerating the industry's structural shift toward downstream and value-added products. Looking ahead, downstream and value-added segments could account for an increasing share of both volume and value in the steel sector, reflecting the industry's strategic pivot toward higher value capture and end-user orientation. The next decade's investment cycles will test whether the sector's financial strengths can match its technological and environmental ambitions.

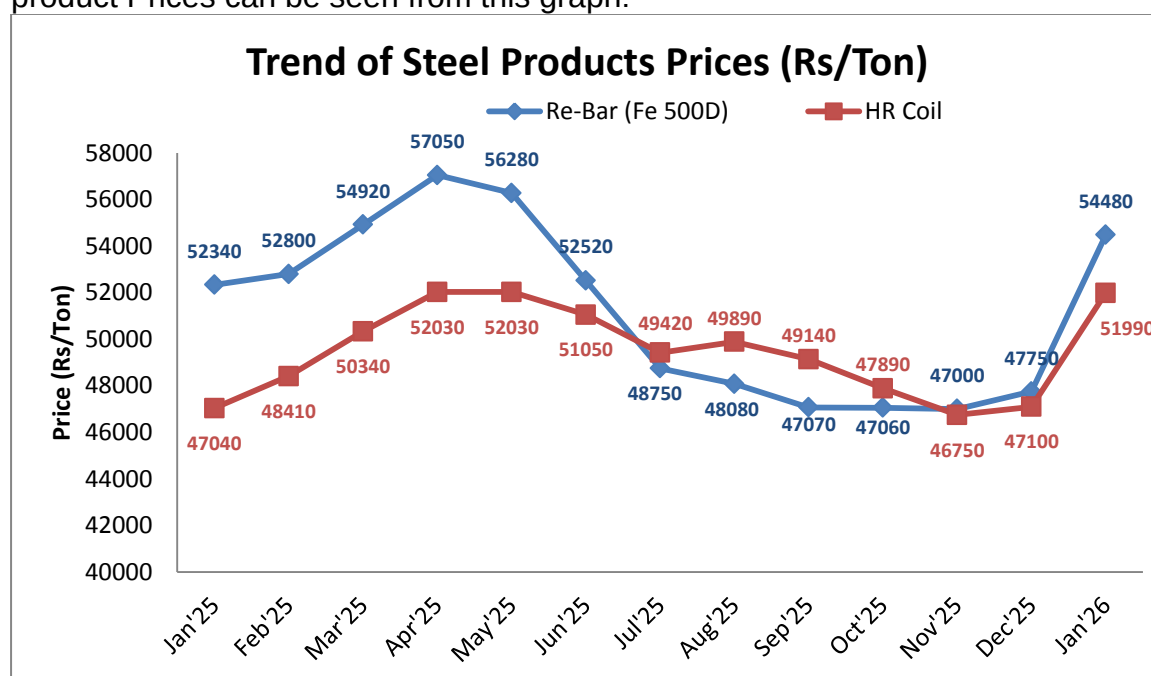
Meeting emission targets could also play a significant role in shaping the domestic industry. In the near term, steel companies are reducing emissions by optimizing energy efficiency and leveraging process improvements in BF-BOF using measures such as installing control systems to reduce coke rate, optimizing slag chemistry setpoints, and increasing BOF gas capture. However, in the long run, several low-emission technologies, such as hydrogen-based direct reduced iron (DRI); carbon capture, use, and storage; and near-zero-emissions electric arc furnaces (EAFs), are being explored globally—but their readiness and cost structures remain uncertain in the Indian context.

Besides transition options, there is growing policy support to meet the 10ecarbonisation goals. India introduced the Carbon Credit Trading Scheme in fiscal

year 2025, the National Green Hydrogen Mission, the revised Production-Linked Incentive 1.1 (PLI 1.1) for specialty steel, and end-of-life vehicle rules (with a target of 18 percent recycled content by fiscal year 2036). However, scaling is likely to demand deeper capital and more stringent transition mandates.

Capacity additions and 11ecarbonisation mandates could become indicators of how India navigates the supply of raw materials. India's iron ore equation is shifting from surplus to uncertainty. Although production of iron ore increased to about 280 million metric tons in 2024, growing demand could lead to an increased reliance on imports by 2035. Global demand for met coal, a critical raw material for BF-BOF steelmaking, is likely to decline by 1 to 2 percent due to falling demand from China, Europe, and the Commonwealth of Independent States. However, with the addition of new BF-BOF-based steel capacities, India could lead the demand for seaborne met coal, growing from 72 million metric tons in 2023 to between 120 million and 130 million metric tons in 2035. As Indian steelmakers pivot toward lower-carbon production routes, steel scrap is emerging as a critical, yet limited, resource. India consumed approximately 33 million metric tons of scrap in fiscal year 2024, of which 30 percent was imported. By 2030, demand is estimated to be 50 million to 60 million metric tons, with a shortfall of 10 million to 15 million metric tons.

Steel Prices : Price regulation of iron & steel was abolished on 16.1.1992. Since then, domestic steel prices are determined by the interplay of market forces. Domestic steel prices are influenced by trends in raw material prices, demand – supply conditions in the market, international price trends among others. As a facilitator, the Government monitors the steel market conditions and adopts fiscal and other policy measures based on its assessment. The fluctuations in Steel product Prices can be seen from this graph:-



Road ahead

- As India advances its industrialization and infrastructure goals, achieving sustainable growth and maintaining global competitiveness could depend on

strategic shifts—specifically, focusing on high-growth sectors, accelerating 12ecarbonisation efforts, securing raw materials, boosting competitiveness, and collaborating with the government.

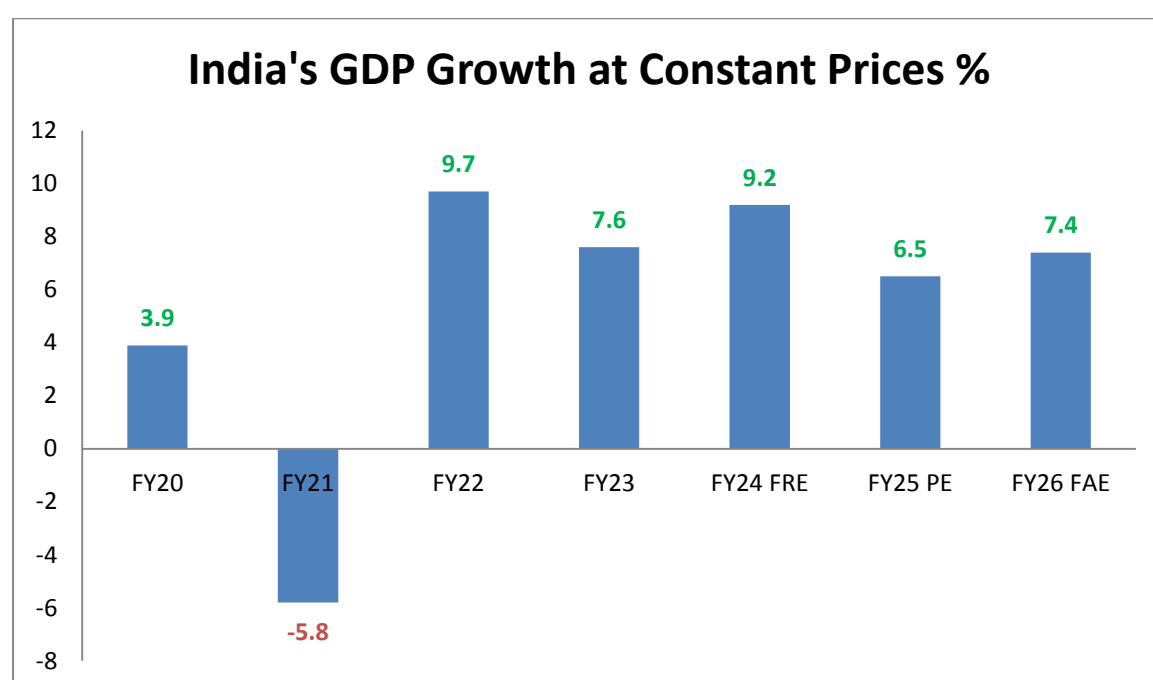
- To meet the evolving demands and ensure business sustainability, Indian steelmakers may need to assess and optimize their portfolios. Their focus will, most likely, turn to high-growth segments of the domestic economy, such as infrastructure, renewable energy, defense, and electric vehicles, while also maintaining access to stable export markets.
- Building on this strategic adaptation, Indian steelmakers may be required to weigh their options toward 12ecarbonisation. The pathways for a low-carbon future include developing a road map for integrating low-carbon technologies such as natural gas and H₂-DRI while enhancing energy efficiency in existing BF-BOF setups, securing long-term access to scrap, green power, and low-emission raw materials. Various funding options—including PLI schemes, concessional finance, and climate-linked instruments to offset 12ecarbonisation capital expenditures—could be explored to chart the path to 12ecarbonisation.
- Looking ahead, steelmakers could reimagine their supply chains to focus on reducing the reliance on imported met coal by focusing on domestic exploration, among other means, and shifting some of their capacity to EAF. They could also address cost-efficiencies by ensuring access to low-cost power and deploying AI for quality control and predictive maintenance.
- Overarching these imperatives is the collaboration with the government to improve logistics, enhance raw material resilience, strengthen quality frameworks, drive 12ecarbonisation, and improve India's competitiveness in high-end export markets.
- Sustained value creation and global differentiation require targeted strategic pivots by steelmakers. Such actions can position India as a leading manufacturing hub and economic driver even amidst evolving geopolitical challenges and sustainability requirements.

1.4 Outlook for Indian Economy

The year 2025 marked an inflection point: Policy overhauls across Western economies—particularly in trade, investment, and industrial policy—triggered spillover effects across all major global markets. India was not immune to these shifts. Intricately connected to global value chains, India, the world's fourth-largest economy and a major global trading partner, faced external shocks and acute effects from these global policy changes, including tariff escalations and volatile capital flows. Yet, despite headwinds, demand resilience, a reset in trade and

investment outlook, and policy reforms stood out. India focused squarely on its biggest strength, domestic demand, to keep growth buoyant as inflation levels stayed low at 1.8% on average through the fiscal year.

Much of 2025 efforts comprised managing external shocks and strengthening domestic fundamentals. Consequently, a major milestone came in August 2025, when S&P upgraded India's sovereign rating from BBB– to BBB—its first such upgrade in 18 years. As India enters 2026, several themes will shape the next phase of growth and demand the same level of pragmatism. We expect full fiscal year growth to be revised substantially upward, as third-quarter numbers are likely to remain strong due to festive spending. Deloitte Projects GDP Growth to stand between 7.5% and 7.8% in fiscal 2025 to 2026, and then between 6.6% and 6.9% in fiscal 2026 to 2027, buoyed by the rollout of new goods and services tax (GST) rules and slowing inflation.



Source: MoSPI

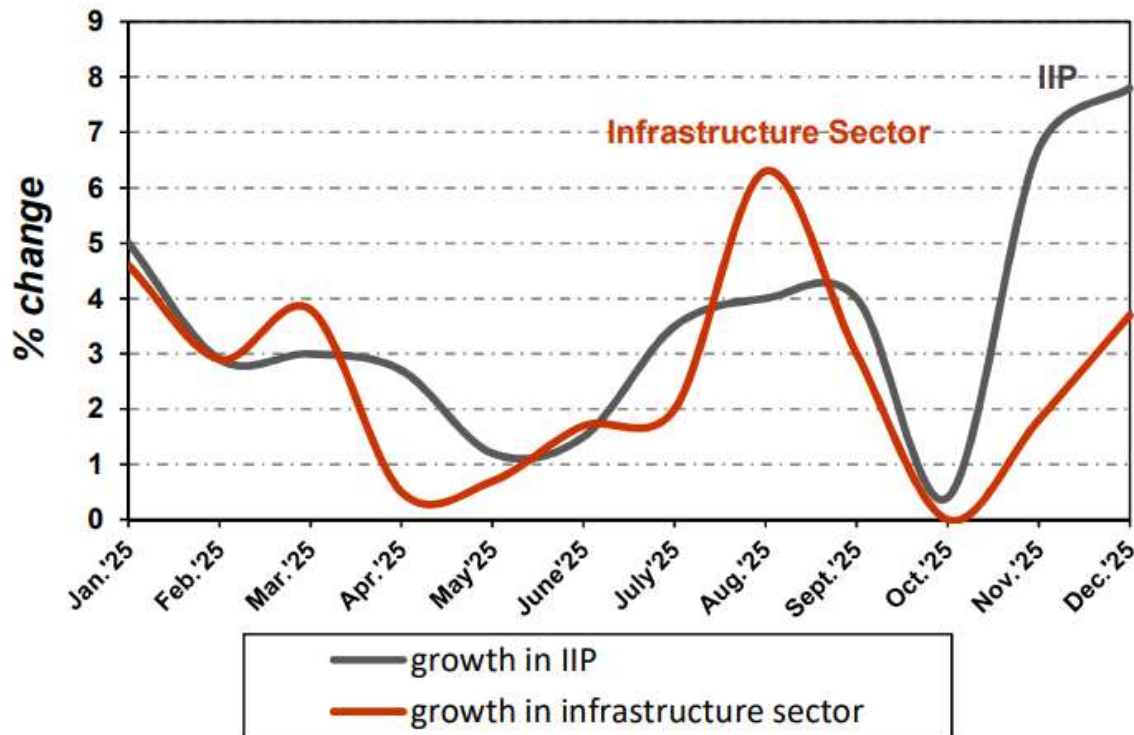
India's GDP growth trajectory reflects strong resilience after the pandemic shock. Following the sharp contraction of -5.8% in FY21, the economy rebounded robustly to 9.7% in FY22 and maintained healthy momentum at 7.6% in FY23 and 9.2% in FY24. Although growth moderates to 6.5% in FY25 (PE) and is projected at 7.4% in FY26 (FAE), the overall trend signals stable expansion with normalization after the high post-COVID base effect. The data underscores India's position as one of the faster-growing major economies, supported by domestic demand and investment momentum.

India's repo rate cycle over the last few years reflects a clear shift from pandemic-era stimulus to inflation control:

- May 2020 – April 2022: Repo rate held at 4.0% to support economic recovery post-COVID.
- May 2022 – Feb 2023: Aggressive tightening phase; rate raised by 250 bps to 6.50% to curb inflation.

- FY24 – FY25: Rate largely maintained at 6.50%, signaling a pause to balance growth and inflation risks.
- Recent stance (2025-26 phase): RBI maintained its key repo rate at 5.25% during its February 2026 meeting, after cutting it by 25 bps at the December meeting, in line with forecasts, amid confidence in a softer inflation outlook and improving growth prospects.

Growth in Industrial Sector



Source: MoSPI

As per the IIP Press Release (Dec 2025), India's industrial momentum strengthened sharply toward year-end. The General IIP growth accelerated to 7.8% in December 2025, the highest in over two years, up from 7.2% in November. This recovery aligns with the sharp rebound shown in the chart for November–December. Importantly, Infrastructure/Construction Goods grew by 12.1% in December 2025, emerging as one of the top contributors to overall IIP growth. This validates the spike seen in the infrastructure curve in the chart and indicates that infrastructure-led demand played a key role in driving the late-year industrial upturn.

With slowing global demand, rising trade frictions, and a delicate domestic consumption environment, India deployed a carefully sequenced set of fiscal, monetary, and trade reforms that not only cushioned the economy but also laid the foundation for future growth.

Key drivers

- Consumption: Private final consumption expenditure grew by 7.9% in the second quarter, supported by the lowest inflation level of 1.7% seen in a decade, rising disposable incomes from tax and GST relief, and better rainfall. Consumption grew by 7.5% in the first half of the fiscal.

- Investment: Government capital expenditure utilization rose to 51.8% in the first half of the fiscal year (versus 37.3% last year), boosting gross fixed capital formation growth to 7.6% (versus 6.7% last year).
- Sectoral strength: Gross value added (GVA) grew by 8.1% in the second quarter of the fiscal year, with manufacturing up 9.1% and services surging 9.2%, led by financial and professional services. GVA growth for the first half of the year came in at around 7.8%. Services now contribute 60% of GVA and 48% of exports, underscoring their strategic role.
- Exports: After a strong first quarter, exports moderated in the second due to higher US tariffs on Indian exports (including a 50% tariff on select goods). But a trade rebound is expected, supported by upcoming trade agreements with the United States and the European Union, as well as diversification into services, electronics, and pharmaceuticals.

High-frequency indicators, including strong consumer confidence, vehicle registrations and sales, and easing inflationary pressures, point to continued support for economic activity. However, some external indicators such as currency depreciation and FPI outflows show emerging signs of stress, posing downside risks to the outlook.

India has consolidated expenses, and the fiscal deficit is targeted at 4.4% of GDP this fiscal year, down from pandemic highs of 9.2% in fiscal 2020 to 2021. Disciplined expenditure management and buoyant revenue streams have, so far, helped government pursue growth-supporting measures.

Going forward, reforms—including the rollout of GST 2.0 and tax-relief measures—are unlikely to derail this trajectory. Higher nontax revenues, driven by an accelerated disinvestment pipeline and strategic asset monetization, are expected to offset potential shortfalls, ensuring fiscal prudence alongside growth.

Forecasts for India GDP growth for FY'25-26

<i>RBI</i>	7.3%
<i>IMF</i>	7.3%
<i>Fitch</i>	7.4%
<i>ADB</i>	7.2%
<i>Deloitte India</i>	7.5-7.8%

1.5 Steel Policies and Recent Initiatives of Ministry of Steel

A. NSP 2017 – Vision, Mission & Objectives

Steel is a product of large and technologically complex industry having strong forward and backward linkages in terms of material flows and income generation. It is also one of the most important products of the modern world and of strategic importance to any industrial nation. From construction, industrial machinery to

consumer products, steel finds its way into a wide variety of applications. It is also an industry with diverse technologies based on the nature and extent of raw materials used.

In India, steel has an output multiplier effect of nearly 1.4 on GDP and employment multiplier factor of 6.8.

Today, the Indian steel industry contributes approximately 2% to the country's GDP and employs about 5 lakh people directly and about 20 lakh people indirectly. The National Steel Policy 2017 (NSP 2017) is an effort to steer the industry to achieve its full potential, enhance steel production with focus on high end value added steel while being globally competitive.

India's competitive advantage in steel production is driven, to a large extent, from the indigenous availability of high grade iron ore and non-coking coal – the two critical inputs of steel production. In addition, it also has a vast and rapidly growing market for steel, strong MSME sector and a relatively young work force with competitive labour costs.

Driven by the positive demand outlook and prevailing high prices of steel in the period post 2004, the Indian steel sector witnessed a wave of investments in the states of Odisha, Jharkhand, Karnataka and Chhattisgarh. Substantial new capacity was created and existing plants were modernized. A significant portion of these investments were funded by banks and other forms of borrowings.

India today is 2nd largest steel producer in the world. There is significant potential for growth given the low per capita finished steel consumption of 82 Kg in India, as compared to world average of 224 Kg. Indian economy is rapidly growing with enormous focus on infrastructure and construction sector. Several initiatives mainly, affordable housing, expansion of railway networks, development of domestic shipbuilding industry, opening up of defence sector for private participation, and the anticipated growth in the automobile sector, are expected to create significant demand for steel in the country. Further, while the main focus of the industry is on the domestic market, being in close vicinity of the developed west and developing east, provides it a strategic location that augurs well for the industry seeking opportunities for exports of finished goods and imports of some scarcely available raw materials.

Over the past two decades, the Indian steel industry has developed capabilities of producing a wide range of value added steel at par with global best practices addressing diverse needs of the end user industries. However, India still needs to make a special effort to domestically produce number of value added products like automotive steel for high end applications, electrical steel (CRGO), special steel and alloys for Power equipment, Aerospace, Defense and Nuclear applications.

The Indian steel sector is disadvantaged due to limited availability of some of the essential raw material such as high grade lumpy Manganese ore & Chromite, coking coal, steel grade limestone, refractory raw material, Nickel, Ferrous Scrap etc. Due to shortage of domestic coking coal, both in terms of quantity and quality, pig iron producers/ BF operators in India have to significantly depend on import of coking coal.

In the recent past, multiple issues have also adversely impacted the steel sector, viz. cancellations of iron ore and coal mine allocations, delays in land acquisition, environmental clearances, which led to many of the projects facing significant cost and time overruns. Additionally, companies also faced substantially increased operating costs on account of increased logistics & raw material costs and other charges.

Vision: To create a technologically advanced and globally competitive steel industry that promotes economic growth.

Mission: Provide environment for attaining –

- i. Self-sufficiency in steel production by providing policy support & guidance to private manufacturers, MSME steel producers, CPSEs & encourage adequate capacity additions.
- ii. Development of globally competitive steel manufacturing capabilities
- iii. Cost-efficient production and domestic availability of iron ore, coking coal and natural gas
- iv. Facilitate investment in overseas asset acquisitions of raw materials.
- v. Enhance domestic steel demand.

Objectives: The National Steel Policy aims at achieving the following objectives –

- i. Build a globally competitive industry
- ii. Increase per Capita Steel Consumption to 160 Kgs by 2030-31
- iii. To domestically meet entire demand of high grade automotive steel, electrical steel, special steels and alloys for strategic applications by 2030-31
- iv. Increase domestic availability of washed coking coal so as to reduce import dependence on coking coal from ~85% to ~65% by 2030-31
- v. To have a wider presence globally in value added/ high grade steel
- vi. Encourage industry to be a world leader in energy efficient steel production in an environmentally sustainable manner.
- vii. Establish domestic industry as a cost-effective and quality steel producer
- viii. Attain global standards in Industrial Safety and Health
- ix. To substantially reduce the carbon foot-print of the steel industry

NSP 2017 aims to increase focus on expansion of MSME sector, improve raw material security, enhance R&D activities, reduce import dependency and cost of production, and thus develop a “technologically advanced and globally competitive steel industry that promotes economic growth” eyeing self-sufficiency in production, developing globally economical steel manufacturing capabilities by facilitating investments and cost-efficient productions with adequate availability of raw materials.

With focus on R&D, the technology would be of utmost focus over the next decade and MSME steel plants would be the key drivers to achieve the additional capacity required for India’s consumption led growth and improvement in the overall productivity and quality.

Expected impact / outcome of NSP 2017 The following targets have been set in the NSP 2017:

S.No.	Parameter	Projections (2030-31)
1.	Total crude steel capacity (in MTPA)	300
2.	Total crude steel demand/production (in MTPA)	255
3.	Total finished steel demand/production (in MTPA)	230
4.	Sponge iron demand/production (in MTPA)	80
5.	Pig iron demand/production (in MTPA)	17
6.	Per Capita Finished Steel Consumption (in Kgs)	158

The other expected impacts are as under:

a) India to be world leader in energy efficiency and sustainability, in association with suitable agency, will constantly monitor techno-economic performance of all the steel plants within the country vis-a-vis the global best practices. Transfer of technology for production of automotive steel and other special steels will be facilitated by JVs with global leaders.

b) 145 Indian Standards for steel and steel products have already been notified under the mandatory quality certification mark scheme of BIS. Efforts will be made to bring in additional steel products, which are used in critical end-use applications, under the mandatory scheme to ensure protection of human health, environment and safety.

c) Attain global standards in Industrial Safety and Health The Ministry is coordinating with steel companies to ensure that on the job trainings on maintaining a safe workplace are provided to employees of the steel companies.

d) Substantially reduce the Carbon footprint of the industry In order to address the environment related issues, the Ministry is facilitating the formation of a forum to chalk out best practices and is also focusing on development of a Waste Management Plan for the industry.

e) Domestically meet the entire demand of high grade automotive steel, electrical steel, special steel and alloys.

Segment wise demand projections :

Segments	Demand (15-16)	Demand (30-31)
Infrastructure (oil refinery, highway, bridges, port, air-port, transportation, urban infrastructure, industrial sheds, pre-fab buildings)	9.5	90
Construction (Real Estate)	23.5	45
Engineering & Fabrication (Capital Goods, Consumer Durables, boilers, General engineering)	35	43
Automotive	2.5	10
Railways	2	5
Packaging (LPG cylinders, grain bins, GI boxes)	2	6
Energy (power projects, wind mills, power transmission)	3	11
Ship Building	4	3
Oil & gas pipeline		4
Defence, space, nuclear		2
Others		11
Total	81.5	230

B. Policy for providing preference to Domestically Manufactured Iron and Steel Products (DMI&SP) Policy in Government Procurement

The Government had introduced DMI&SP Policy on 8th May, 2017 to provide preference to domestically produced iron and steel material in Government tenders. Further, to fine tune this objective, the Policy was revised on 29th May, 2019 and on 31st December, 2020. The salient features of the Policy are as under

- This policy provides preference to Domestically Manufactured Iron and Steel Products (DMI&SP) in Government procurement.
- The policy covers a list of 49 manufactured products of iron and steel. The policy also covers capitals goods for manufacturing iron and steel products.
- While earlier the domestic content was specified as 15-50 per cent on the 49 products of iron and steel, the new list of 49 products have minimum prescribed value addition ranging between 20-50 per cent making it difficult for imported steel to compete with domestic bidders for government contracts.
- Each Ministry or Department of Government and all agencies/entities under their administrative control is under the purview of the DMI&SP policy as notified by the Ministry of Steel. All Central Sector Schemes (CS)/Centrally Sponsored Schemes

- (CSS) for which procurement is made by States and Local Bodies come within the purview of this Policy, if that project / scheme is fully / partly funded by Government of India.
- The policy is applicable to projects where the procurement value of iron and steel products is greater than Rs.5 lakh. The policy is also applicable for other procurements (non-project), where annual procurement value of iron and steel products for that Government organization is greater than Rs. 5 lakh. However, it shall be ensured by procuring entities that procurement is not split for the purpose of avoiding the provisions of this policy.
- The policy is applicable to purchase of iron and steel products by private agencies for fulfilling an EPC contract and/or any other requirement of Ministry or Department of Government or their CPSEs and also to capital goods for manufacturing iron and steel products in compliance to prescribed quality standards, as applicable.
- No Global Tender Enquiry (GTE) shall be invited for tenders related to procurement of iron and steel products. No Global Tender Enquiry (GTE) shall be invited for tenders related to procurement of Capital Goods for manufacturing iron and steel products having estimated value upto Rs. 200 Crore except with the approval of competent authority as designated by Department of Expenditure.
- The policy has provisions for waivers to all such procurements, where specific grades of steel are not manufactured in the country, or the quantities as per the demand of the project cannot be met through domestic sources.

The policy is envisaged to promote growth and development of domestic steel Industry and reduce the inclination to use low quality and low cost (unfairly traded) imported steel in Government funded projects.

Sl. No.	Plant shop	Capital goods	Domestic value addition requirement
1	Raw material handling system	Apron feeder, barrel couplings, heavy-duty bearings, hydraulic disc brakes, tanker & container for powdered materials, conveyor belt for pipe conveyors, high angle conveyor system, crushers, crane rail lubrication system, four girder EOT Crane, crane weighing system, crane air conditioning, fluid couplings, forklift trucks, hydraulic motors, hydraulic system, locking-assembly (friction grip), load cells, level sensors, pipe conveyor system, plough/paddle feeder, pneumatic transportation – dense & lean phase, reclaimers, radio remote control, rail fixing arrangements(special), rapid/ flood loading system, stackers, special screen, slew ring bearings, tippers, transfer cars, tongs (special), vibration, isolation system(spring damper), wagon tippers, wagon loaders	50%
2	Mineral beneficiation (iron ore and coal) equipment	Industrial crushers, grinding mills, conventional screens, slurry pumps, hirate thickeners, filters, hydro clones	50%
3	Coke oven	Coke Oven Silica Refractory, Anchorage System, Waste gas valve with branch pipe, Flash Plate, Door Frame, door body, Minor Casting: Gooseneck, Valve box, AP Lid, Charging & inspection hole lid and frame Reversing mechanism, Centralised lubrication system, Hydrojet Door Cleaning Mechanism, Spillage code conveyor system, skip hoist, Door Lowering Rack, Isolation/Reversing Cocks, Level II automation, Oven machines	50%
4	By-product plant	Primary Gas Cooler, Electrostatic Tar Precipitator, H ₂ S, NH ₃ & Naphthalene Scrubber, Combi Stripper, Flushing Liquor Pump, Claus Kiln, Claus reactors, Waste Heat Boilers, Decanters	50%

5	Sinter Plant equipment	Pallet car, Drive/discharge end Sprocket assembly, Curved rail, Slide rails, Hot sinter breaker and Grizzly, Dip rail & running rail, Impeller assembly for Process fan, Drive assembly of Sinter machine, Hi-intensity Mixer & Noduliser	50%
6	Pellet plant equipment	Pallet car, Drive/discharge end Sprocket assembly, Curved rail, Slide rails, running rail, Vertical roller mill, Impeller assembly for Process fan, Drive assembly of Indurating machine, Hi-intensity Mixer, Balling disc, Single deck roller screen and Double deck roller screen	50%
7	Blast furnace equipment	Bellies stop system with Bleeder valve, SG Iron stave coolers, Copper stave coolers, Stock level indicator (Radar Type), Mud gun, Drilling machine and Manipulator, Gas Cleaning Plant system, Top Recovery Turbine system including its by-pass valve, De-bricking Machine, Re-railing equipment, PCI system, Grinding mill for PCI, Stock level indicator, Tuyere Stock assembly, Waste Heat Recovery system, BF & Hot Blast Stoves Technological Valves, Above Burden probes, Slag granulation unit, Tuyere&Tuyere cooler, Torpedo Ladle Car, BF hearth refractory	50%
8	Direct reduction plant equipment	Charge distributor, Upper & lower seal leg, Reformer & Recuperator system, Burden feeders, Turbo-expander, Process Gas Compressor, Seal gas compressors & bottom seal gas compressors, Seal gas generators & driers, Process Gas Heater, CO ₂ removal plant	50%
9	Basic oxygen furnace equipment	Main and Maintenance equipment comprising of converter, gunning machine, Refractory/slag monitoring device, converter vessel, trunnion ring and suspension system, trunnion bearings and housing, Converter bull gear unit and tilt drive system, Rotary joint for converter, bottom stirring system, Lance body with clamping, Lance copper tips, Valve stations for oxygen blowing/ bottom stirring, Sub-lance system, Off gas analyzer with process module i.e. Process software/ hardware, container lab Measurement probes, Switch over station, ID fan for primary gas, Hot metal and steel ladle, Ladle Transfer car, Ladle maintenance equipment, Slag pot, Slag pot transfer car, Scrap boxes, Scrap Transfer car, Lance carriage, Lance guide, Crane & hoist, Lance hoist & trolley, Lance tilting device, Traverse for lifting lances, Bunker of various sizes, Bin Vibrator, Weighing Hopper, Maintenance stands, De dusting suction hood, Teeming/HM, ladle relining stands, Stand Cooling stack inspection device, Hood traverse carriage, Refractories, Bypass & isolation valves, Flare stack & ignition system, Scrubbing tower shell - Wet gas cleaning system, Dog house, Ladle drier, ladle pre-heater, ladle cooler, Fume collection hoods, Clean gas stack, Dust silo, Weigh Bridge, Slag retaining device	50 %
10	Electric arc furnace	Furnace proper (includes furnace lower shell, upper shell and roof, Tilting platform, Furnace Gantry) and transformer, Electrode regulation system, Hydraulic system, Refractories, Parts of Level I & Level II Automation system. LF - water cooled ladle roof, electrode mast and arms, electrode regulating system, wire feeding system, Bottom inert gas stirring Valve stand for porous plug and top lance, Emergency lance mechanism, Lance carriage system with drive unit, Automatic temperature, sampling & bath level / O ₂ measurement, Temp. & oxygen immersion lance, lance carriage system with drive unit, Hydraulic system, Refractories, Ladle roof Delta portion, RH proper (includes Ladle transfer car, vacuum vessel, Vessel lifting & lowering system. Hydraulic system, Multi Function lance, Valve racks/station, Electrode clamp unit, conductor of electrode arms, water cooled cable, A R stirring valve rack, lance transport car, Refractory lance, Hydraulic cylinder, Ladle roof lifting cylinder, Lubrication system, Suction hood, damper, Vibro feeder, weighing hopper, wire feeding system, Electrode nipping stand, Cranes, hoist, Temperature & sampling tips, ladle stands, ESP, Deducting hoods, Refractories, bag filter, Cranes etc.	50%

11	Continuous casting equipment	Ladle turret, ladle cover manipulator, Ladle Shroud manipulator, tundish car, Continuous tundish temperature measurement system, Tundish stopper rod mechanism, emergency cut-off gate, mould assembly, Nozzle quick change device, mould oscillator and EMS system, Electro-Magnetic braking system, Strand guide segment, Withdrawal & Straightening unit (WSU), Roll gap checker, Emergency torch cutter, Torch cutting machine, Deburrer, Marking machine, Technological control system & process models, Black Refractories, strand guide segment, tundish, ladle cover, roller tables & auxiliaries, mould & segment maintenance equipments, tundish maintenance equipments, EMBR system	50%
12	Flat product mills	Large castings and forgings like mill housing, bed plates, work rolls, backup rolls, end spindles; roller tables, backup roll and work roll chucks, coilers / tension reels / uncoilers, AGC cylinders, shears, levelers, laser welders, packaging machines, non-contact gauges / profile gauges, anti-friction roll neck bearings, oil film bearings, gear boxes, mill motors	50%
13	Long product mills	Mill housing, bed plates, work rolls, backup rolls, spindles; roller tables, coilers/tension reels / uncoilers, shears, billet welder, packaging machines, non-contact gauges / profile gauges, anti-friction roll neck bearings, oil film bearings, finishing blocks, gear boxes, mill motors	50%

Impact of the DMI & SP Policy

The increased domestic value addition is expected to contribute to the vibrant steel sector and the associated industries by generating employment and domestic market for their products. This policy has provided and expected to provide significant savings to the Indian Economy and restrict the use of low quality and cheap imported steel in Government funded projects, alongside developing domestic capability for import substitution. DMI&SP Policy has so far resulted in import substitution of Rs. 26,600 Crore approximately.

C. Steel Import Monitoring System (SIMS) for import data dissemination

Steel Import Monitoring System (SIMS) has been institutionalized which is an online platform for advance registration of intended imports of steel in order to provide granular data on steel imports, 0-60 days in advance to help the Ministry and the industry identify the exact grade being imported into the country in order to plan domestic manufacturing, besides giving advance warning about any surge in imports. SIMS platform was launched on 16th September 2019 for import consignments started at the Port of Entry w.e.f. 1st November 2019. SIMS registration is fully online and automated without any human intervention and registration number can be obtained by the steel importer after making an online payment of token registration fee prescribed for this purpose. SIMS has enabled the domestic industry to plan their pricing and production strategy and helped the country move towards Aatmanirbhar Bharat in steel making.

D. Quality Control Orders/BIS

Government has been facilitating supply of quality steel for critical end-use applications such as infrastructure, construction, housing and engineering sector.

Ministry of Steel is the leading Ministry with maximum coverage of products under the BIS certifications marks scheme. A total of 145 Indian Standards on Steel and Steel products have been covered under Mandatory Quality Control Orders. These orders prohibit, import, sale and distribution of substandard steel products. The imposition of QCO is in the public interest or for the protection of human, animal or plant health, safety of the environment, or prevention of unfair trade practices, or national security as stated in the BIS Act, 2016. Through the aforementioned orders, Ministry of Steel has so far covered 99 Carbon Steel, 44 Stainless Steel and Alloy Steel products standards and 2 Ferro Alloys under the mandatory BIS Certification Scheme. Further, to meet the requirement of containers manufacturing, Indian Standard 11587 which was already under the purview of quality control order was revised by BIS by including the Corten Steel and the domestic steel manufacturers were urged to apply for BIS certification for the product. Four domestic manufacturers have already been certified by BIS and domestic manufacturers are ready to supply the desired quality of corten steel required by container manufacturers to reduce the dependency of import of corten steel and make container manufacturing industry Aatmanirbhar. In addition, as per the data of imported steel grades shared with BIS, more than 250 new steel grades have been included in the existing standards and 5 new standards are under preparation. This exercise is facilitating the upgradation of the Indian Steel Standards at par with Global Standards. This exercise is also facilitating indigenization of many of the imported steel grades for import substitution and “Make in India” initiative. QCO notified by Ministry of Steel and also the experience gained from the discussions held with the stakeholders have resulted in several benefits which are highlighted below:

- Strengthening of the Indian Standards and also formulation of new standards by BIS based on the information provided by Ministry of Steel.
- Promote indigenisation of the imported steel grades by bringing in the importers and the domestic steel producers together.
- Preventing unfair trade practices such as misdeclaration and under-invoicing of the imported steel consignments. Based on the information shared by Ministry of Steel, the relevant authorities have imposed measures such as ADD, penalties for misdeclaration, etc.

The Ministry of Steel's recent clarification regarding the enforcement of 151 BIS (Bureau of Indian Standards) Standards under the Quality Control Orders (QCOs) marks a pivotal move towards ensuring uniform quality in steel products across the value chain. While the government has not issued any new QCO after August 2024, its June 13 order emphasizes that intermediate steel materials used in the manufacturing of BIS-standard final products must also conform to BIS norms. This move is set to be a game-changer in curbing substandard imports and levelling the playing field for domestic steelmakers—but it also raises concerns for India's micro, small, and medium enterprises (MSMEs), which are integral to the steel industry ecosystem. India's steel market has increasingly been flooded with cheaper, substandard imports, especially in the wake of global overcapacity and declining steel consumption in countries like China. The government's clarification seeks to address a critical disparity: Indian steelmakers must use BIS-compliant intermediate materials, whereas importers have had more relaxed standards for finished goods. This loophole not only undermines the quality of products but also places Indian manufacturers at a competitive disadvantage. By mandating that HR/CR coils and

other intermediate inputs must also adhere to BIS standards when used in downstream products like coated steel, the government is rightly strengthening the integrity of the steel supply chain. This will improve product reliability, boost consumer confidence, and reduce safety risks in infrastructure, construction, and automotive sectors.

The directive effectively restricts the use of non-BIS (Bureau of Indian Standards) input, previously sourced from smaller re-rollers/processors or through imports, tightening compliance requirements for producers. The changes are already causing concern among small- and mid-sized processors, many of whom rely on non-certified inputs. The regulatory changes are set to impact the sourcing behavior for steel processing and trading companies, while also acting as a non-tariff based barrier for such firms thus constraining imports further (safeguards duty already in place), reversing domestic steel prices to some extent depending on the quality of law enforcement and implementation, favoring large integrated steel mills like Tata Steel, JSW, AM&NS in the process given limited availability of BIS-compliant semis in the domestic market and added hurdles for imports. Demand for BIS-certified sheets, slabs, billets, blooms and wire rods is likely to grow. Uncertified rerollers, processors and stockists may face pressure to adapt. While such regulatory shifts may cause short-term supply deficits, elevated input costs for these processors/traders, there are hopes that the overall steel market (including secondary players) moves towards greater degree of standardization (through certification process institutionalized over a period of time), reducing lead times for the sustainable steel journey and enhancing steel exports potential to international markets.

E. Production Linked Incentive (PLI) Scheme

PLI Scheme for domestic production of specialty steel has been approved with an outlay of Rs. 6322 crore by the Cabinet. The five broad categories of Specialty steel, identified under the scheme, are used in a variety of applications including white goods, automobile body and components, pipes for transportation of oil and gas, boilers, ballistic and armour sheets, high-speed railway lines, turbine components, distribution and power transformers. The Scheme has been notified on 29.07.2021 and detailed Scheme Guidelines were published on 20.10.2021. The application process through online system was available from 29.12.2021 till 15.09.2022. The scheme is set to commence from FY:2023-24 (PLI to be released in FY:2024-25). 57 MoUs have been finalized out of 67 applications from 30 companies which were selected under the Production Linked Incentive (PLI) Scheme for Specialty Steel. This will attract committed investment of Rs. 29530 crore with a downstream capacity addition of 25 million tonne and employment generation potential of 70000

PLI Scheme 1.1

2nd Round of application under PLI Scheme (gazette Notification dtd. July 29, 2021) Launched on January 6, 2025,



Minimum Qualifying Thresholds of PLI 1.1 Scheme:-

Sl. No.	Sub-category	Investment	Capacity	Min y-o-y incremental rate (%)
		(in ₹ Cr)	(in '000 tonnes)	
1	Galvanneal/GI-Auto-Gr	700	400	10
2	Tin mill Products	600	200	20
3(a)	Coated/Plated products of Metallic/Non-Metallic alloys	200	250	10
3(b)	Al-Zn coated (Galvalume)			30
4	Colour Coated	300	250	20
5(a)	HR Coil, Sheets and Plates API Gr 52<=X<=70	2,750	4,500	25
5(b)	HR Coil, Sheets and Plates API Gr>X-70			20
5(c)	High Tensile Sheets, Coil, Plates, YS>=450			10
6	Auto Gr Steel AHSS (CRCA: Cold rolled closed annealed)	1,000	900	15
7(a)	Boiler Quality, Pressure Vessels	2,500	1,200	15
7(b)	QT/Abrasion Resistance and Wear Resistance			30

Source: MoS PLI Document

Tenure & Permissible Investment in PLI 1.1:-

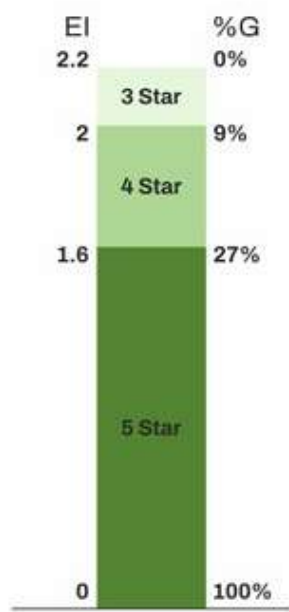
- 5 years
- Production period : FY 2025-26 to FY 2029-30
- Incentive disbursement period : FY 2026-27 to FY 2030-31
- Investment made after 6th January, 2025 shall be eligible.

Provisions for PLI Scheme 1.1

- Capacity addition- Enhancement made by applicant during scheme tenure by installation of new unit(s) under the applied product sub-category
- Capacity augmentation- Enhancement made by applicant during scheme tenure by augmenting existing facility(ies) in production capacity and/ or product quality under the applied product sub-category

F. Green Steel Taxonomy- India

MoS proposes star rating
for public procurement
of 'green steel'



- “Green Steel” shall be defined in terms of percentage greenness of the steel, which is produced from the steel plant with CO₂ equivalent emission intensity less than 2.2 tonnes of CO₂e per tonne of finished steel (tfs). The greenness of the steel shall be expressed as a percentage, based on how much the steel plant’s emission intensity is lower compared to the 2.2 t-CO₂e/tfs threshold.
- Based on the greenness, the Green steel shall be rated as follows:
 - Five-star green-rated steel: Steel with emission intensity lower than 1.6 t-CO₂e/tfs.
 - Four-star green-rated steel: Steel with emission intensity between 1.6 and 2.0 t-CO₂e/tfs.
 - Three-star green-rated steel: Steel with emission intensity between 2.0 and 2.2 t-CO₂e/tfs.

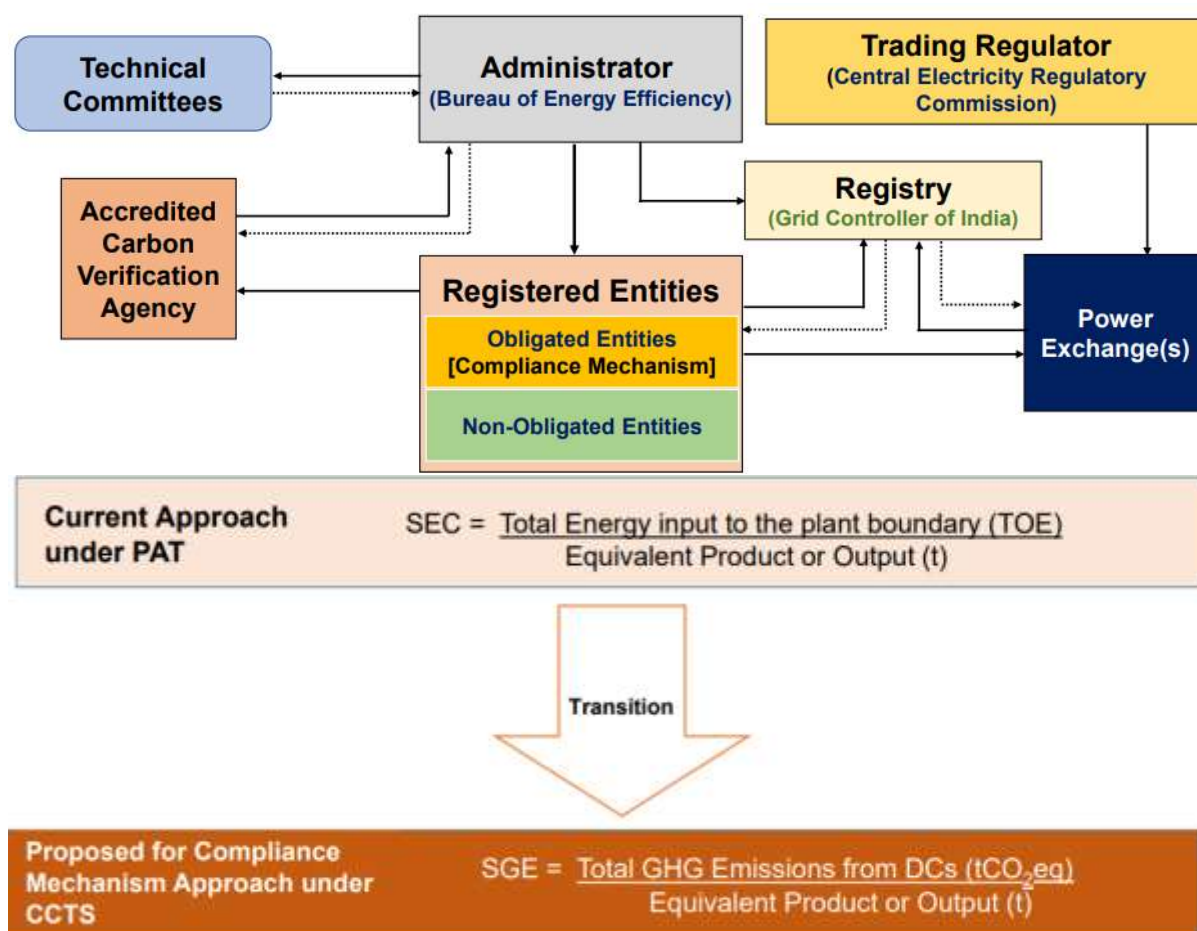
Steel with emission intensity higher than 2.2 t-CO₂e/tfs shall not be eligible for green rating.

- The threshold limit for defining the star rating of Green Steel shall be reviewed every three years.
- The scope of emissions shall include Scope 1, Scope 2, and limited Scope 3, up to finished steel production. Scope 3 emissions shall include agglomeration (including sintering, pellet making, coke making), beneficiation, and embodied emissions in purchased raw materials and intermediary products, but shall not

include upstream mining, downstream emissions and transportation emissions, both within and outside the gates of a steel plant.

- The National Institute of Secondary Steel Technology (NISST) shall serve as the nodal agency for measurement, reporting, and verification (MRV) as well as for issuing the greenness certificates and star ratings for the steel.
- The certificate shall be issued on yearly basis (financial year). In case the steel plants opt for MRV more frequently, then the certificate may be issued more than once in a year as per the requirement.

G. India Carbon Market



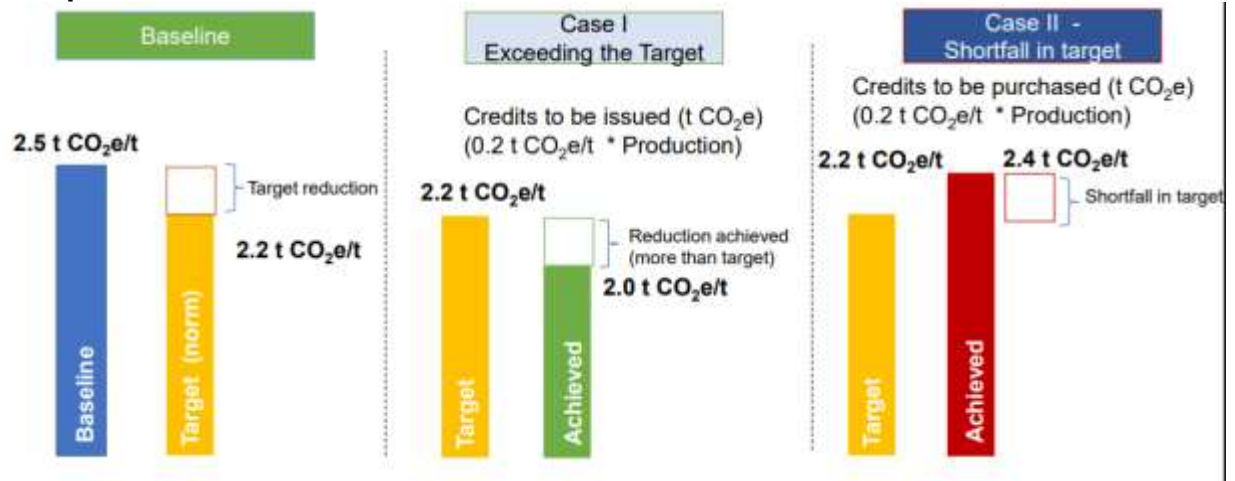
India's carbon market for the steel sector is part of the national Carbon Credit Trading Scheme (CCTS), a framework established in 2023 to reduce greenhouse gas emissions. The CCTS includes a compliance mechanism for energy-intensive industries like steel, setting mandatory emission intensity targets that steel companies must meet. Companies that reduce their emissions can generate carbon credits, while those that exceed their targets may need to purchase credits to comply.

Key components of the carbon market for steel

- **Mandatory compliance:** The CCTS mandates that companies in the iron and steel sector meet specific greenhouse gas (GHG) emission intensity targets, defined as the amount of CO₂ emitted per tonne of crude steel produced.
- **Compliance market:** This is a mandatory program where regulated entities must comply with their set emission targets.

- Voluntary offset mechanism: The CCTS also includes a voluntary market where non-obligated entities can register projects to reduce, remove, or avoid emissions to earn carbon credit certificates (CCCs).
- Trading platform: Carbon Credit Certificates (CCCs) will be issued on an electronic registry and can be traded on an electronic trading platform.
- Incentivizing decarbonization: The scheme is designed to incentivize steel companies to reduce their carbon footprint and invest in cleaner production methods.

Compliance Mechanism in Indian Carbon Market



H. Carbon Border Adjustment Mechanism



Carbon Border Adjustment Mechanism is a unique carbon price-based measure, which has been evolved by the European Union (EU) alone. No other economic region has implemented any similar measure till date. It is to take effect from 2026, and will replace the existing Emission Trading Scheme (ETS), the latter being based on CO₂ free allowances. The levy will align with the EU's carbon market price and prevent carbon leakage. The carbon price will be paid in the country of production of the imported goods. It is based on the principle of equal carbon pricing, ie, EU businesses currently pay a carbon price in order to have the allowance to meet CO₂ emission norms, based on their production in the EU.

I. PM GatiShakti National Master Plan

With the help of Bhaskaracharya Institute for SpaceApplications and Geoinformatics (BiSAG-N) the infrastructure Ministries have uploaded their rail, road, port networks,

etc. on PM GatiShakti National Portal. Ministry of Steel has onboarded itself on PM GatiShakti Portal (National Master Plan portal) with the help of a mobile application created by BiSAG-N, by uploading the Geo locations of more than 2100 (Twenty one hundred) steel units (including big players) functioning in the country. The Geo location of all the Iron ore Mines and Manganese ore mines has also been uploaded. Ministry of Steel is in the process of uploading the geo locations of the existing slurry pipelines and the laboratories functioning in the steel sector. In addition, Ministry of Steel, in line with the goal of PM GatiShakti Master Plan, has identified 22 high impact projects to develop multimodal connectivity and bridge the missing infrastructure gaps. Planned expansion of railway lines, creation of new inland waterways, roads, ports, gas pipeline connectivity will result in creating much needed logistics solution which will drive the steel sector towards achieving its targeted goals by 2030-31, as delineated in NSP 2017.

J. Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act)

The MMDR Act, 1957 is the principal legislation governing the exploration, extraction, and management of mineral resources in India. It lays down the legal framework for granting mineral concessions, regulating mining operations, and ensuring sustainable and transparent development of the mineral sector. For iron ore miners, the MMDR Act defines how leases are obtained, renewed, transferred, and monitored, ensuring that mining contributes to both industrial growth and environmental stewardship.

Key Provisions of MMDR Act

1. Ownership and Regulation

- The State Governments are the owners of minerals within their territories, but all leases and operations are regulated under the MMDR Act and rules framed by the Central Government.
- The Central Government (MoM) provides overarching policy direction and oversight.

2. Mineral Concessions- The Act provides for three types of mineral rights:

- Reconnaissance Permit (RP): For initial survey and exploration.
- Prospecting Licence (PL): For detailed exploration to establish mineral reserves.
- Mining Lease (ML): For actual extraction and production of minerals.

For iron ore, mining leases are granted only through competitive auction.

3. Auction-Based Allocation

- Since the 2015 amendment, auction is the only mode of granting mining leases for iron ore and other major minerals.
- This ensures transparency, fair valuation, and maximum revenue to the State.
- The successful bidder signs a Mine Development and Production Agreement (MDPA) before starting operations.

4. Duration and Renewal

- A mining lease for iron ore is valid for 50 years.
 - There is no provision for renewal; after expiry, the lease area is re-auctioned.
 - Captive and non-captive distinctions have been abolished to promote flexibility in mineral utilization.
5. Regulation of Production and Dispatch
 - Miners must adhere to production caps and environmental conditions stipulated in the approved Mining Plan (prepared as per MCDR 2017).
 - All production, sales, and dispatches must be reported through the Indian Bureau of Mines (IBM) using the MCDR online reporting system.
 6. Mineral Conservation and Sustainable Mining
 - The Act mandates systematic and scientific mining, reclamation of mined-out areas, and zero-waste utilization of minerals.
 - IBM and State DMGs conduct inspections to ensure compliance.
 - Violations can lead to suspension of operations or cancellation of the lease.
 7. Revenue and Levies- Iron ore miners must pay:
 - Royalty (as per rates notified by the Central Government) & Dead rent,
 - District Mineral Foundation (DMF) contribution (10–30% of royalty), and
 - National Mineral Exploration Trust (NMET) contribution (2% of royalty).
 8. Transfer and Sale
 - Leases obtained through auction are transferable with government approval, enabling consolidation and investment inflow.
 - Sale of ore must comply with the conditions of the mining lease and the Mineral (Other than Atomic and Hydro Carbons Energy Minerals) Concession Rules, 2016.

K. Environment (Protection) Act, 1986

Iron ore & Coal mining, due to its extractive nature, directly affects land, water, air, and biodiversity. Under the EPA, mining operations must ensure that environmental standards and safeguards are strictly implemented throughout the mine's lifecycle from exploration to closure.

Key Provisions Applicable to Miners

1. Environmental Clearance (EC)
 - Mandatory under the EIA Notification, 2006 issued under the EPA.
 - Mining projects (≥ 5 hectares) require prior EC from the Ministry of Environment, Forest and Climate Change (MoEFCC) or State Environment Impact Assessment Authority (SEIAA).
 - The EC process involves an Environmental Impact Assessment (EIA) and public consultation.
2. Consent to Establish / Operate

- Mines must obtain consent under the Air (Prevention and Control of Pollution) Act, 1981 and Water (Prevention and Control of Pollution) Act, 1974, enforced via the EPA framework.
- 3. Pollution Control Measures
 - Control of dust emissions from drilling, blasting, crushing, and haul roads.
 - Management of mine water discharge, tailings, and overburden dumps to prevent soil and water contamination.
 - Noise control from heavy machinery and blasting operations.
- 4. Waste and Overburden Management
 - Disposal and stabilization of overburden dumps and tailings must follow environmental safeguards to prevent erosion, siltation, or contamination of nearby streams.
- 5. Forest and Wildlife Protection
 - Activities near protected forests, wildlife sanctuaries, or eco-sensitive zones require additional clearances under Forest (Conservation) Act, 1980 and Wildlife (Protection) Act, 1972, monitored under the EPA framework.
- 6. Environmental Monitoring and Reporting
 - Periodic environmental audits and submission of half-yearly compliance reports to MoEFCC/SEIAA.
 - Mines must maintain environmental quality within limits prescribed under EPA Rules, 1986 (for air, water, noise, soil).
- 7. Mine Closure and Rehabilitation
 - Mandatory progressive and final mine closure plans to restore ecology, vegetation, and land usability.
 - Reclamation of mined-out areas is monitored under EPA provisions.

Steps for Environmental Clearance (EC) :

- Stage I - Screening
- Stage II - Scoping
- Stage III - Public Consultation
- Stage IV - Appraisal

L. Forest (Conservation) Act, 1980

The Forest (Conservation) Act, 1980 (FCA) was enacted by the Government of India to check indiscriminate diversion of forest land for non-forest purposes, including mining, infrastructure, and industrial projects. Its main aim is to ensure ecological balance by making prior approval of the Central Government mandatory before any forest land can be used for activities other than forestry. Iron ore deposits in India are often located in forest-rich regions such as Odisha, Jharkhand, Chhattisgarh, and Karnataka. Mining in such areas involves diversion of forest land, removal of vegetation, and potential ecological disturbance. Therefore, operations fall directly under the purview of the FCA.

Key Provisions Applicable to Miners

1. Prior Central Government Approval
 - Any use of forest land for mining, infrastructure, or beneficiation facilities requires prior approval from the Ministry of Environment, Forest and Climate Change (MoEFCC).
 - This applies to both surface and underground mining operations.
2. Diversion of Forest Land
 - The proposal for diversion must include:
 - Detailed forest area maps & Tree enumeration,
 - Compensatory afforestation plan, and
 - Environmental management measures.
 - Proposals are first reviewed by the State Government and then submitted to MoEFCC for clearance.
3. Compensatory Afforestation (CA)
 - For every hectare of forest diverted, an equivalent area of non-forest land (or degraded forest land twice the area) must be afforested at the miner's cost.
 - CA is mandatory and is monitored through the Compensatory Afforestation Fund Management and Planning Authority (CAMPA).
4. Net Present Value (NPV) Payment
 - Miners must pay the Net Present Value (NPV) for the forest land diverted, representing the ecological and environmental cost of forest loss.
 - The rate varies (₹5–10 lakh per hectare) depending on forest density and type.
5. Wildlife and Biodiversity Safeguards
 - If mining areas fall within or near protected areas (sanctuaries, national parks, or eco-sensitive zones), additional clearances are required under the Wildlife (Protection) Act, 1972.
 - The FCA ensures that biodiversity impact assessments are considered before granting approvals.
6. Monitoring and Compliance
 - MoEFCC, along with Regional Offices and State Forest Departments, regularly monitors compliance with FCA conditions such as:
 - Demarcation of leased forest land,
 - Prohibition on working in unapproved areas,
 - Protection of wildlife corridors, and
 - Implementation of afforestation and reclamation plans.
7. Restrictions on Tree Felling and Land Use Change
 - Felling of trees, removal of vegetation, or construction within forest land without FCA clearance is strictly prohibited.
 - Any violation is treated as a forest offence

M. Engagement with Secondary Steel Sector:

A major segment of iron and steel industry is the segment of secondary producers which contributes more than 40% to the production of crude steel. The role of secondary steel sector in infrastructure development is immense. Not only does infrastructure development provide a stimulus to steel demand but steel intensive construction also leads to rapid building up of infrastructure. Considering the importance of this sector, which mostly consists of MSMEs, Ministry of Steel has

organised a seminar under the chairmanship of Hon'ble Steel Minister with the aim of providing a platform to players in the Secondary Steel sector to share their views on the challenges faced by the sector and ways in which the Ministry can create an ecosystem in which the industry can thrive. The issues raised during the discussion were taken up with concerned Ministries such as Ministry of Finance, Ministry of Port, Shipping and Waterways, Ministry of Coal, Ministry of MSMEs and Ministry of PNG. Ministry of Steel also organized seminars at Bhuvneshwar, Indore, Roorkee and Surat to interact with secondary steel producers and consumers to enhance steel demand in the country.

N. Steel Prices:

Certain measures were taken by the Government to provide relief from high prices of crucial raw materials and intermediates, which included iron and steel. Accordingly, modifications were made in tariffs on raw materials of steel and other steel products vide notification dated 21.05.2022 whereby Import duty on Anthracite/Pulverized Coal Injection (PCI) coal, Coke and Semi-coke and Ferro-Nickel were reduced to zero. Export duty on Iron ores/ concentrates and iron ore pellets was raised to 50% and 45% respectively. In addition, 15% export duty was imposed on pig iron and several steel products.